

Problem

You are given an array of N integers representing the weights of people.

Your task is to divide the people into two groups such that:

- The two groups have the same number of people, or their sizes differ by at most one. The absolute difference between the total weights of the two groups is minimized.

Print the minimum possible difference between the total weights of the two groups.

Submissions

Input Format

The first line contains an integer N, the number of people.

The second line contains N space-separated integers representing their weights.

Leaderboard

Constraints

$2 \leq N \leq 30$ $1 \leq arr[i] \leq 1000$

Output Format

Print a single integer representing the minimum possible difference between the total weights of the two groups.

Discussions

Sample Input 0

```
4
3 4 5 6
```

Sample Output 0

```
0
```

Sample Input 1

```
5
1 2 3 4 5
```

Sample Output 1

```
1
```

```

1  # Enter your code here. Read input from STDIN
2  from itertools import combinations
3  n=int(input())
4  a=list(map(int,input().split()))
5  s=sum(a)
6  k=n//2
7  ans=s
8  for c in combinations(a,k):
9      ans=min(ans,abs(s-2*sum(c)))
10 print(ans)
11
    
```

Line: 11 Col: 1

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

Input (stdin)

Download

```

1  4
2  3 4 5 6
    
```

Your Output (stdout)

```

1  0
    
```

Expected Output

Download

```

1  0
    
```